quandrantal versor [**10**, p. 36]. His attempt at a more basic introduction, *Elements of Quaternions*, ended up being even longer than the *Lectures*.

Quaternions were popularized by Hamilton's disciple Peter Tait, who fought a protracted battle with Josiah Willard Gibbs and Oliver Heaviside, promoters of the rival vector analysis. If quaternion multiplication lacked commutativity, the dot product and the cross product of vectors seemed to have even greater problems, and Tait branded vector analysis a "hermaphrodite monster" [**10**, p. 185].

Telling his great discovery story to his son 23 years after the fact, Hamilton added that at his *eureka* moment he could not "resist the impulse — unphilosophical as it may have been — to cut (the quaternion formula) with a knife on a stone" of a nearby bridge [**17**, 2:435]. Whether this was true or Hamilton simply could not resist the impulse to embellish his story, we will never know, as Time has long erased the etching off the bridge stone — but not off the annals of mathematics. Today the site of Hamilton's vandalism is marked by a plaque which reads, "Here as he walked by on the 16th of October 1843 Sir William Rowan Hamilton in a flash of genius discovered the fundamental formula for quaternion multiplication $i^2 = j^2 = k^2 = ijk = -1$ & cut it on a stone of this bridge". The bridge is now a pilgrimage site for mathematicians seeking to ignite their imagination off the sparks of the original flash.